

FinFETs, Nanosheets and Silicon-on-Insulator

KKKC6353 Nanoelectronic Devices · Lecture 4, Part 2 of 3

Dr Muhammad Faiz Bukhori

Universiti Kebangsaan Malaysia · 2026

The drain reaches under the gate

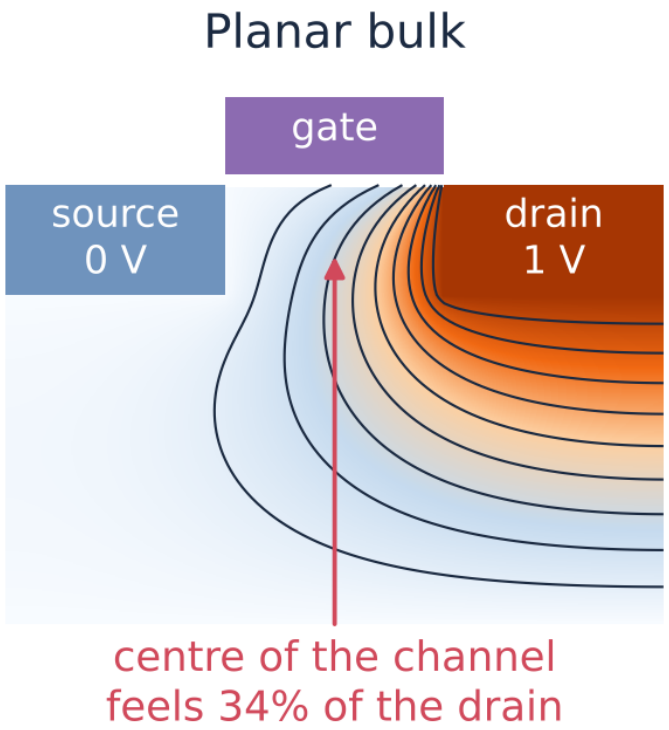


Figure made for this course. Colour: the potential in the silicon with the drain at 1 V (lines every 0.1 V), computed from Laplace's equation for a 20 nm gate.

In a short planar transistor the drain's voltage spreads through the deep silicon under the gate. The middle of the channel feels **a third** of it, so the gate is no longer in charge.

Cure 1: take away the deep silicon

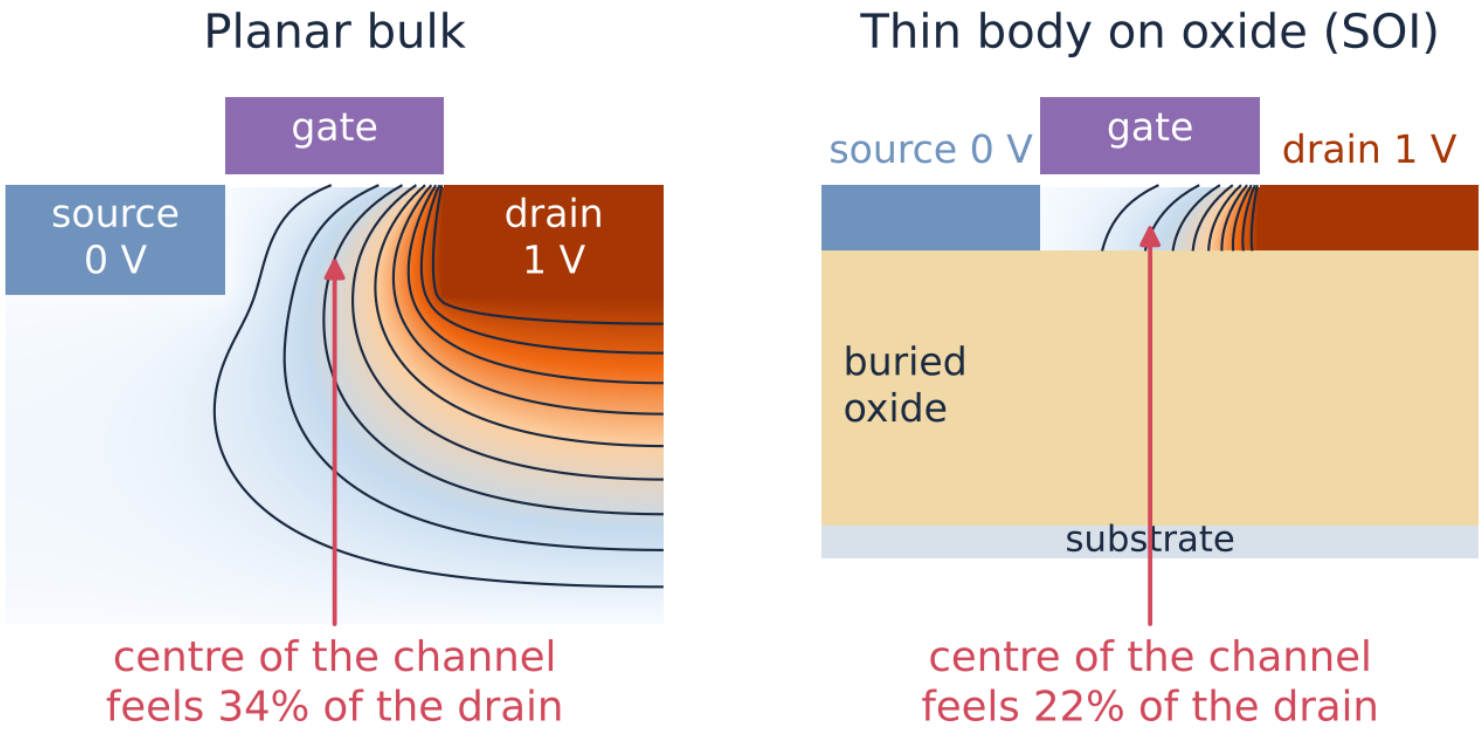


Figure made for this course. Colour: the potential in the silicon with the drain at 1 V (lines every 0.1 V), computed from Laplace's equation for a 20 nm gate.

Build the channel in a **thin film on oxide**: there is no deep silicon for the drain's field to spread through. This is silicon-on-insulator (SOI).

Cure 2: surround the channel with gate

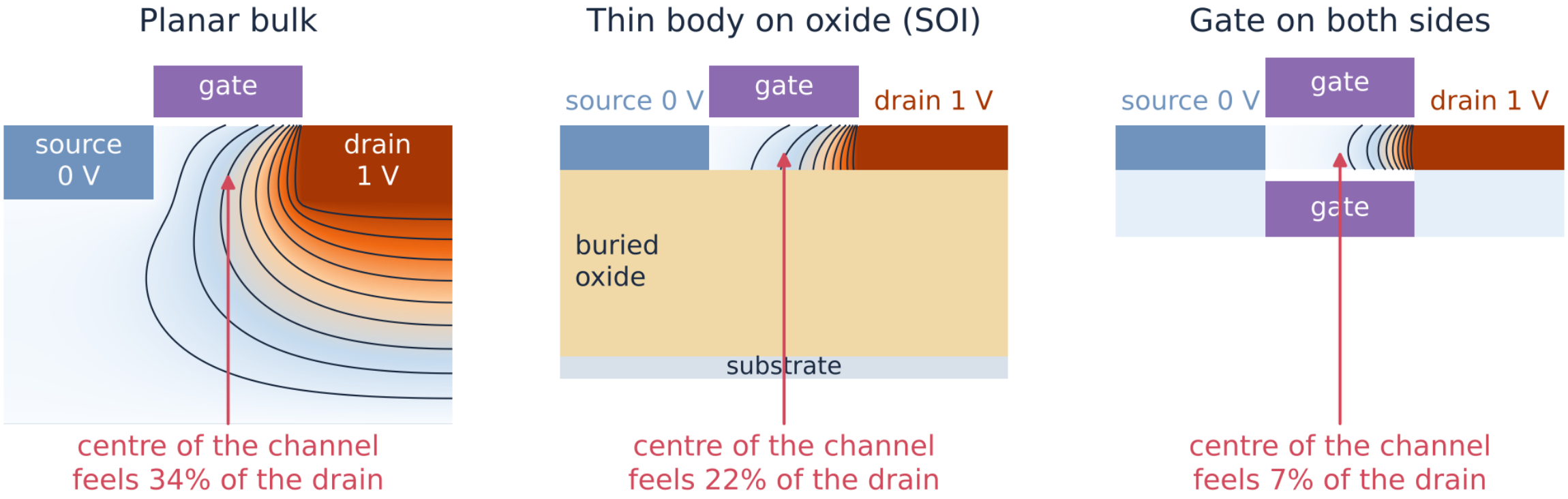


Figure made for this course. Colour: the potential in the silicon with the drain at 1 V (lines every 0.1 V), computed from Laplace's equation for a 20 nm gate.

Gates on **both sides** cut the drain's reach to **7%**. Thin bodies and wrap-around gates are the whole idea behind SOI, FinFETs and nanosheets.

1

Wrapping the gate: FinFETs and nanosheets

Stand the channel on its edge, then stack it.

Planar: one flat channel

- The gate sits **on top** only
- Current flows in a thin layer just under the gate
- More current means a **wider** transistor, and more chip area

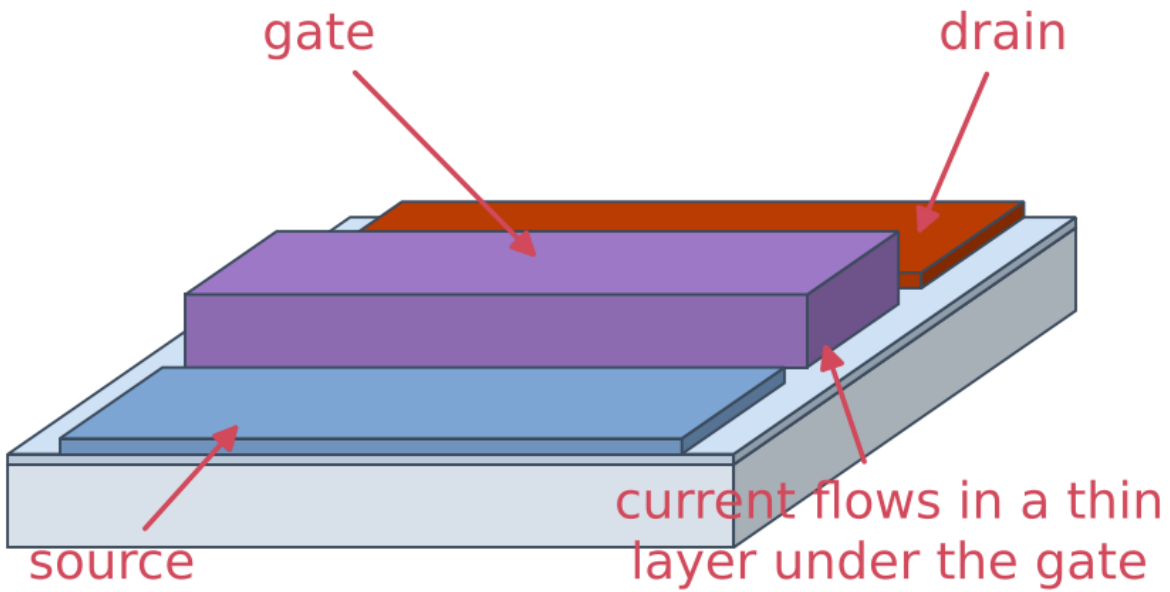


Figure made for this course.

FinFET: stand the channel on its edge

- Thin vertical **fins**, the gate draped over them
- The gate controls each fin from **both sides and the top**
- Current flows down both sidewalls: a **tall** fin carries a lot of current on a small footprint
- Width comes in **whole fins**

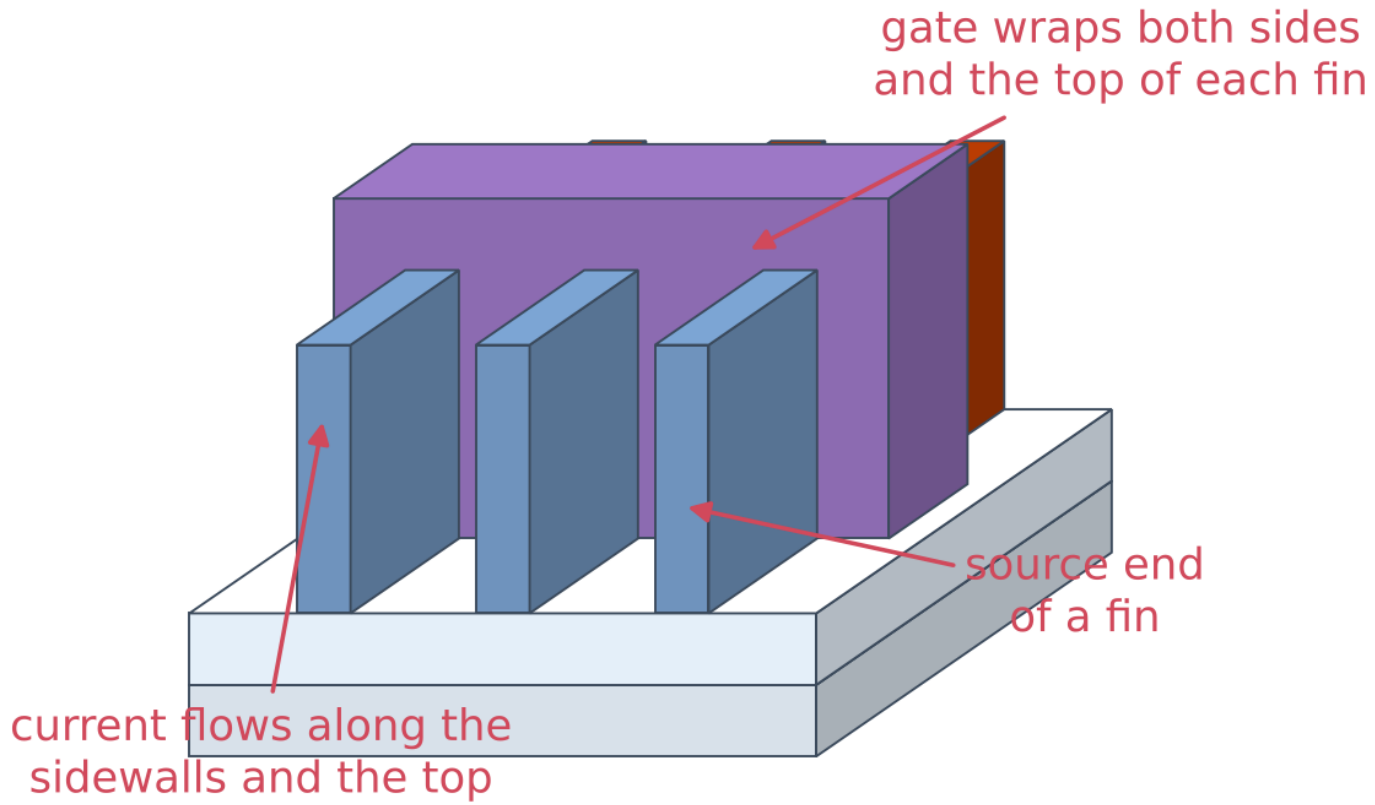


Figure made for this course.

Nanosheet: stack the channels

- Thin silicon **sheets** stacked on top of each other
- The gate fills the gaps: every sheet is gated on **all four sides**
- Sheet width can be **chosen freely**, not in whole fins
- In production since Samsung's 3 nm (2022); TSMC and Intel followed in 2025

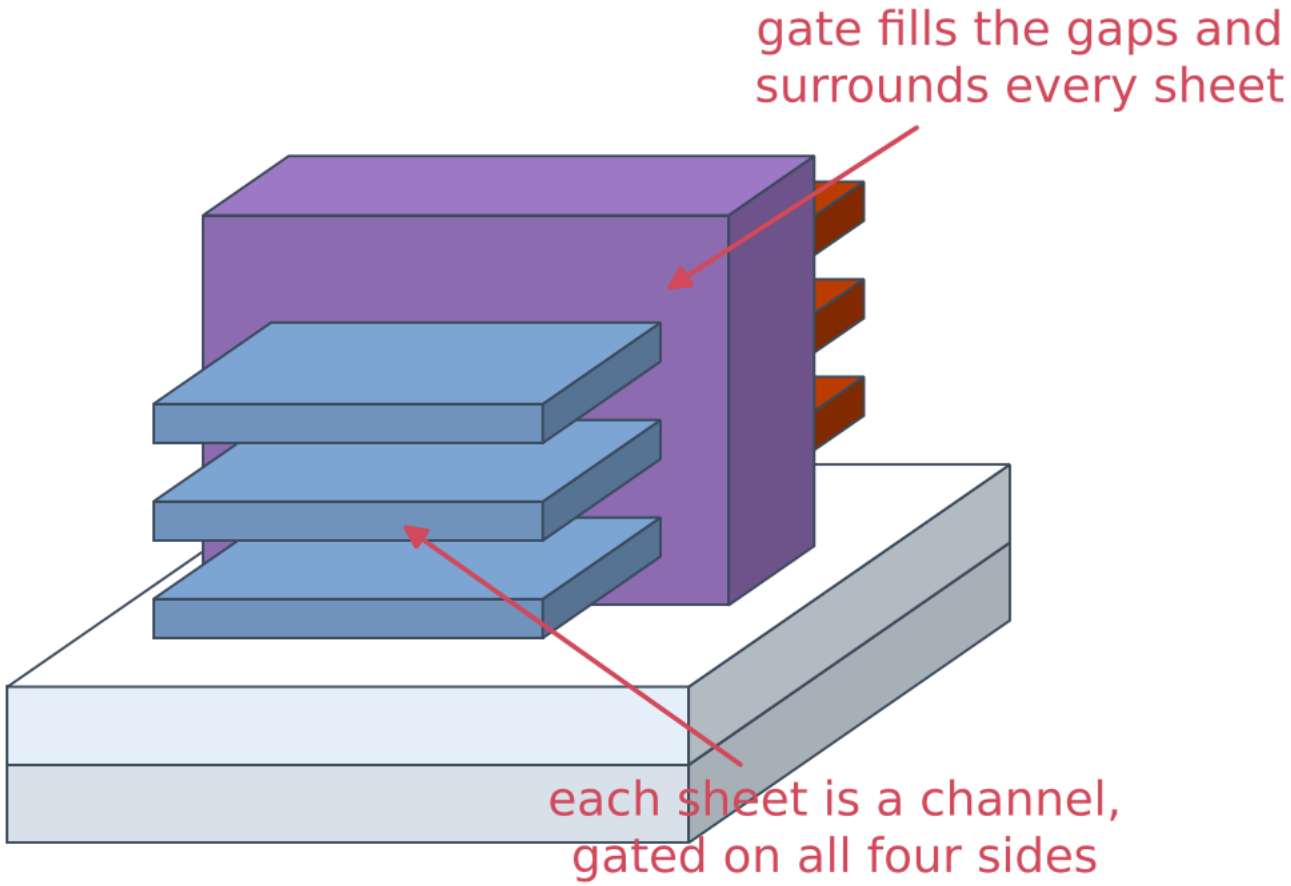


Figure made for this course.

Why fins got taller

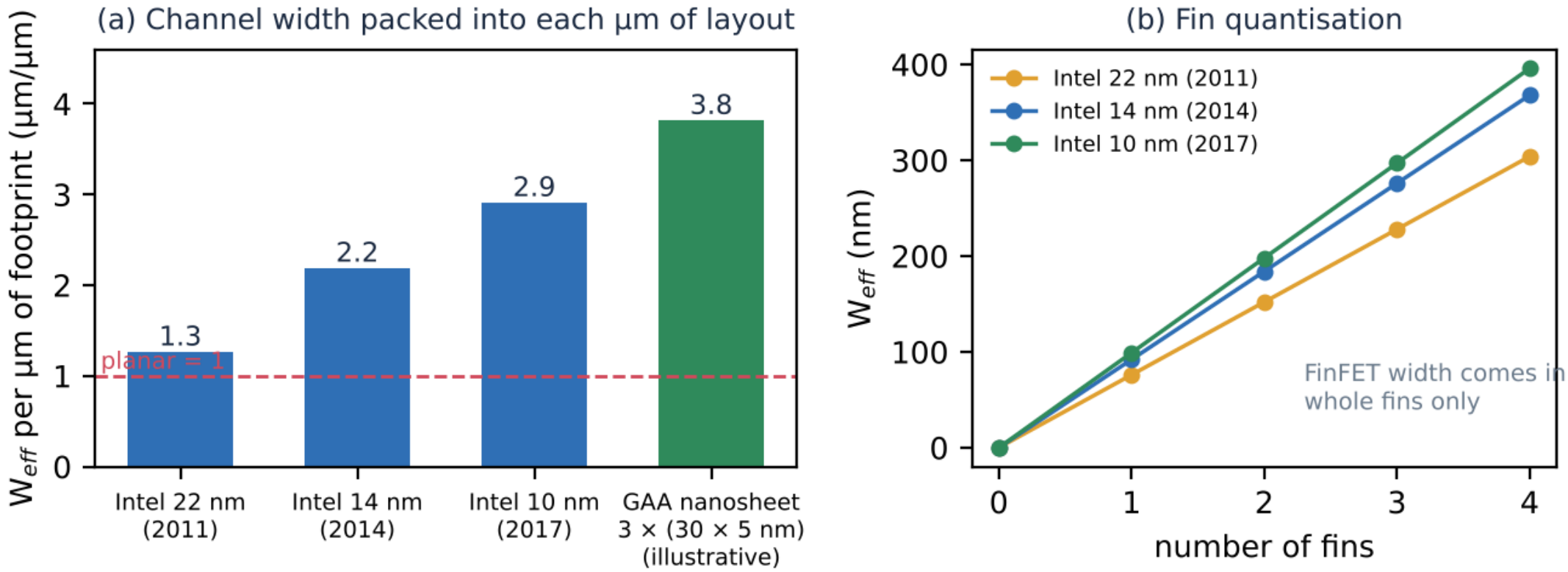
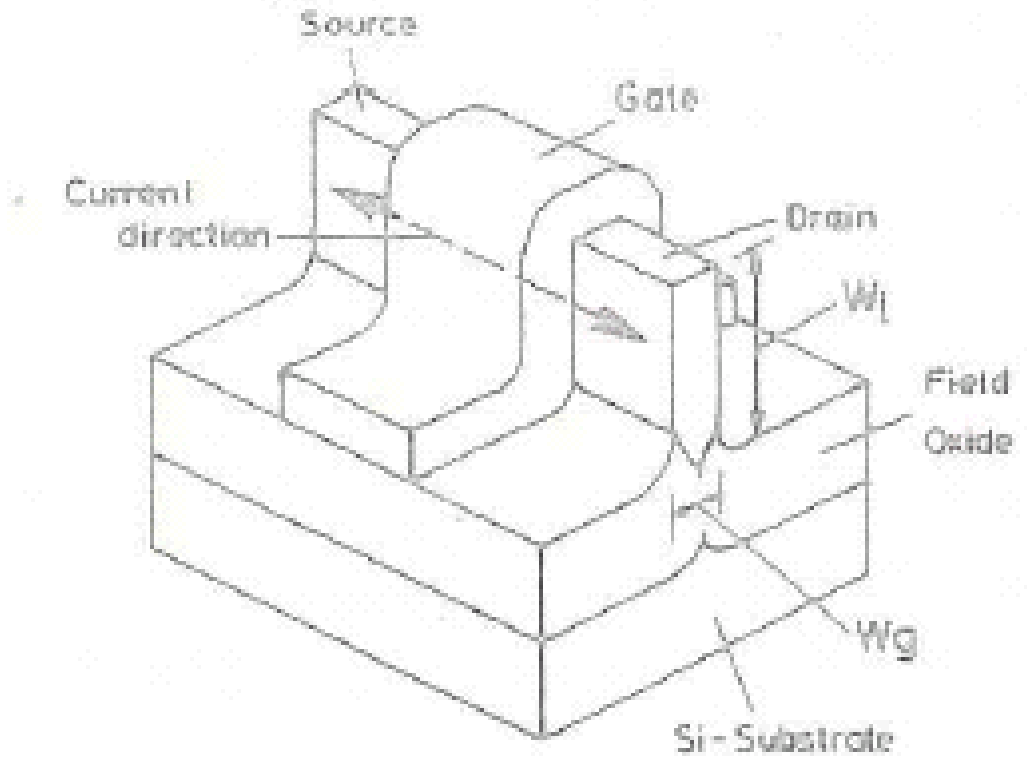


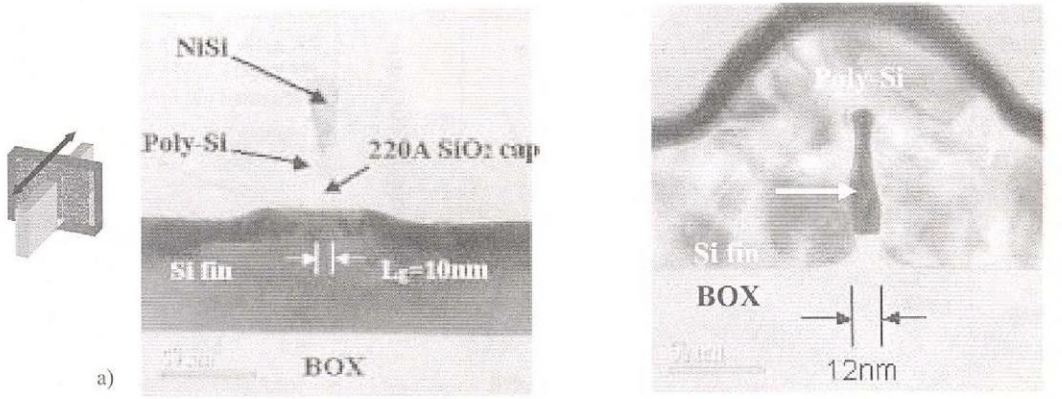
Figure made for this course; Intel fin dimensions from the IEDM papers.

Each generation made fins **taller and closer**: more channel per µm of chip, so **fewer fins** per transistor, which saves area and capacitance.

From the lab to every phone



Hitachi's DELTA: Hisamoto et al., IEEE EDL 11, 36 (1990); reproduced from Q. Zhang, EE 666 slides (2005).



Berkeley FinFET, 10 nm gate: Yu, Hu et al., IEDM 2002; reproduced from Q. Zhang, EE 666 slides (2005).

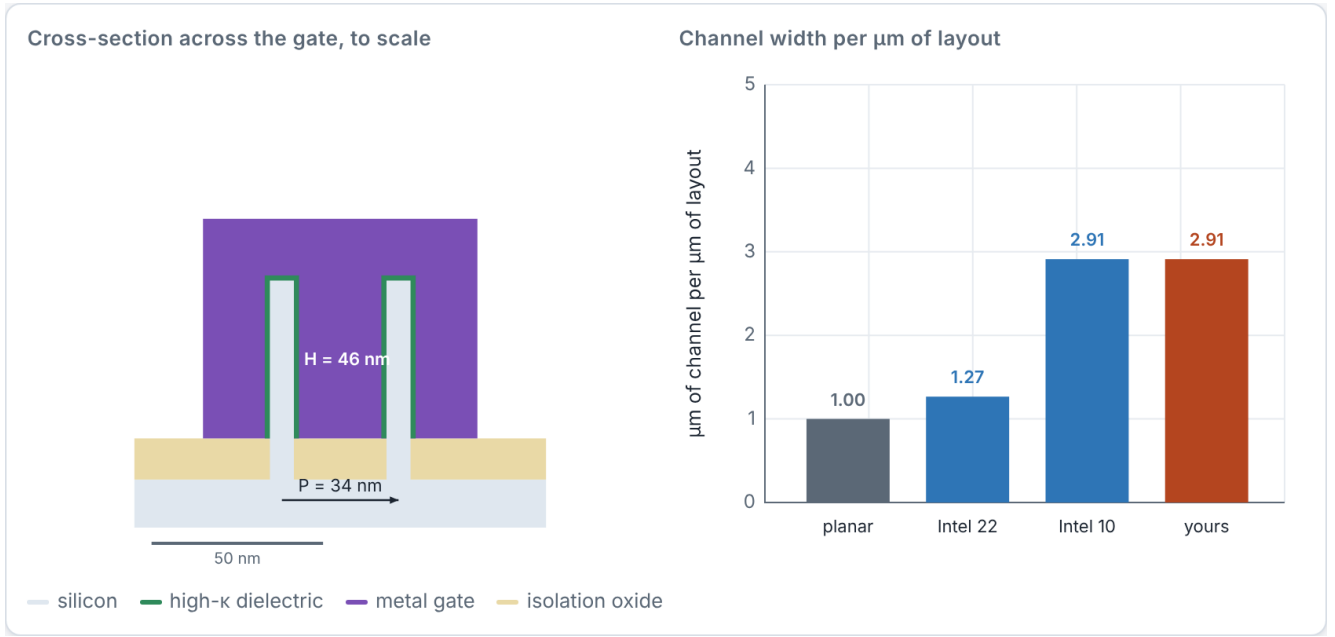
1989 Hitachi wraps a gate over a silicon wall. **1999** Berkeley names it the FinFET. **2011** Intel ships it at 22 nm. **2022–25** nanosheets arrive.

Interactive animation 4

FinFET and nanosheet: width and control

faizverse.github.io/nanoelectronic-devices/L4/finfet-nanosheet.html

- Load the three Intel presets: what changed most?
- Make the fins taller: how many fins do you still need?
- Switch to a nanosheet and change the sheet width
- More detail: thin a 12 nm fin to 6 nm and watch the leakage



Fins thinner than lithography can print

- Print a line, coat it with a thin film, etch back: thin **spacers** stay on its sides
- Remove the line: **two fins** for every printed line
- The fin width is the **film thickness**, set to the ångström

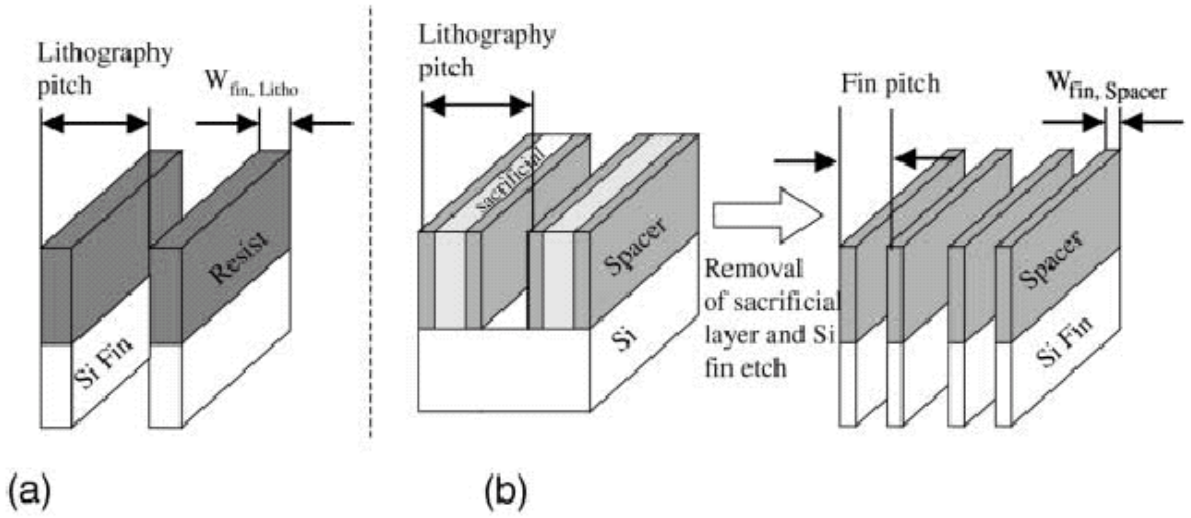
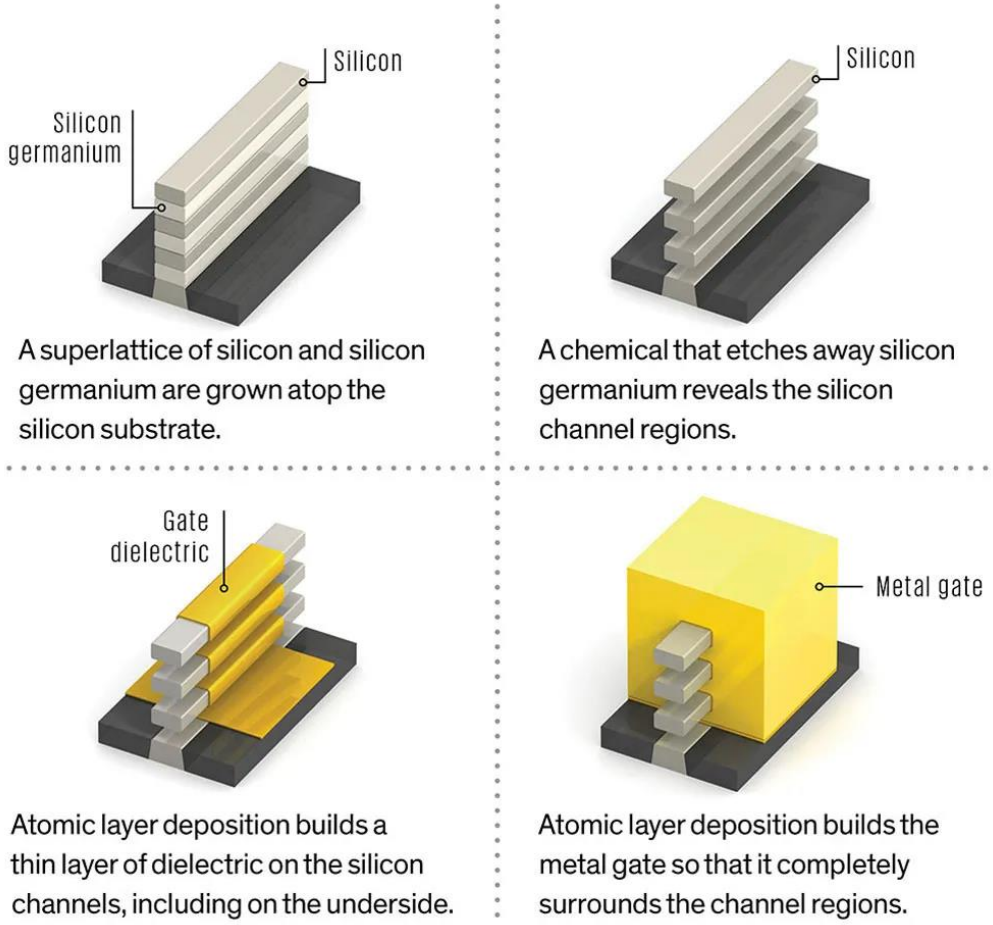


Fig. 3. Comparison of fin density achieved with (a) conventional lithography and (b) spacer lithography technology. The spacer lithography technology produces twice the fin density of a conventional lithography technology.

After Choi et al. (2002) and Rösner et al. (2004); reproduced from Q. Zhang, EE 666 slides (2005).

Making a nanosheet

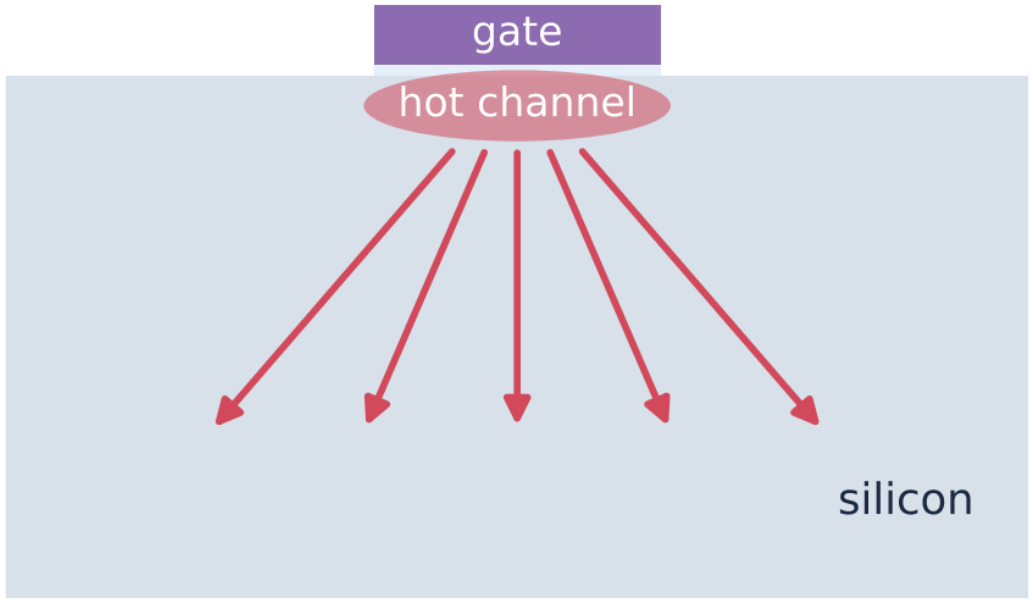
- Grow alternating layers of **silicon and SiGe**
- Etch the SiGe away: the silicon sheets **hang free**
- Coat every sheet with gate oxide and metal, one atomic layer at a time



P. Ye, T. Ernst and M. V. Khare, IEEE Spectrum (2019); from the 2026 course FinFET slides.

The price of three dimensions: heat

Bulk: heat escapes into the silicon



SOI or nanosheet: oxide traps the heat

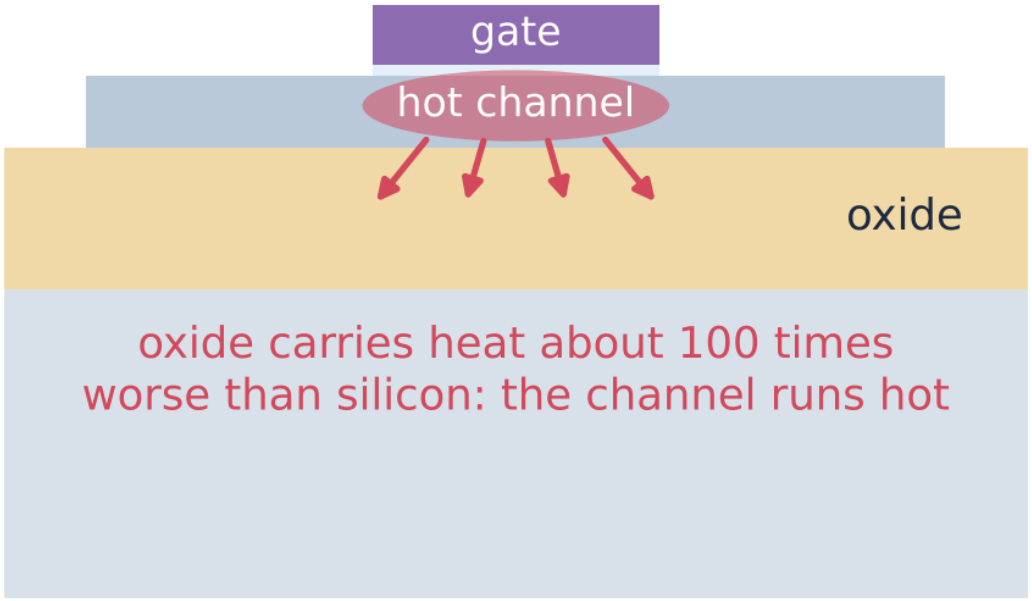


Figure made for this course.

Wrapping the channel in oxide also wraps it in a **heat insulator**: the channel runs hot, which slows it and ages it. Resistance and stray capacitance also grow.

2

Silicon-on-insulator and capacitance

Put every transistor on a layer of glass.

What an SOI wafer is

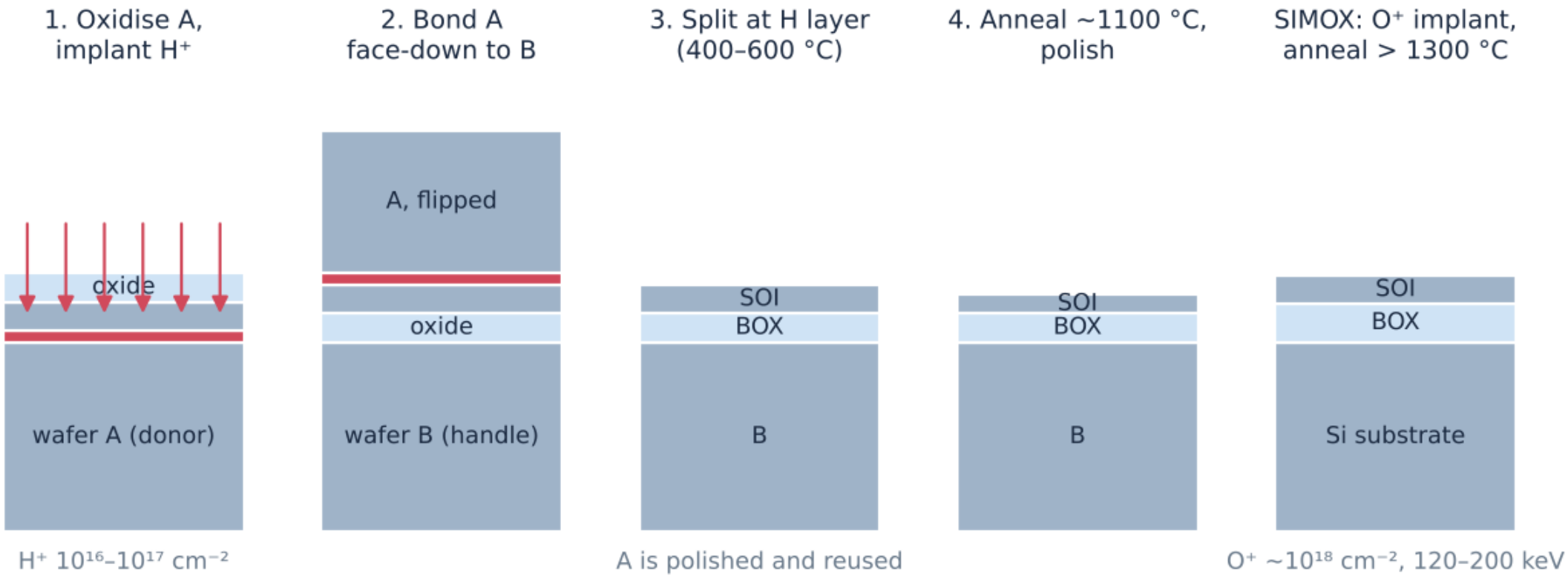


Figure made for this course.

A thin silicon film on a **buried oxide (BOX)** on an ordinary wafer. Smart Cut: implant hydrogen, bond face-down, heat, and the wafer **splits** along the hydrogen layer.

Every drain is a capacitor

- The n^+ drain and the p substrate are both **conductors**
- The depletion layer between them has no free carriers: an **insulating gap**
- So the junction is a **capacitor**, mostly through the drain's large **bottom**
- Every time the output switches, this capacitor must be **charged and discharged**

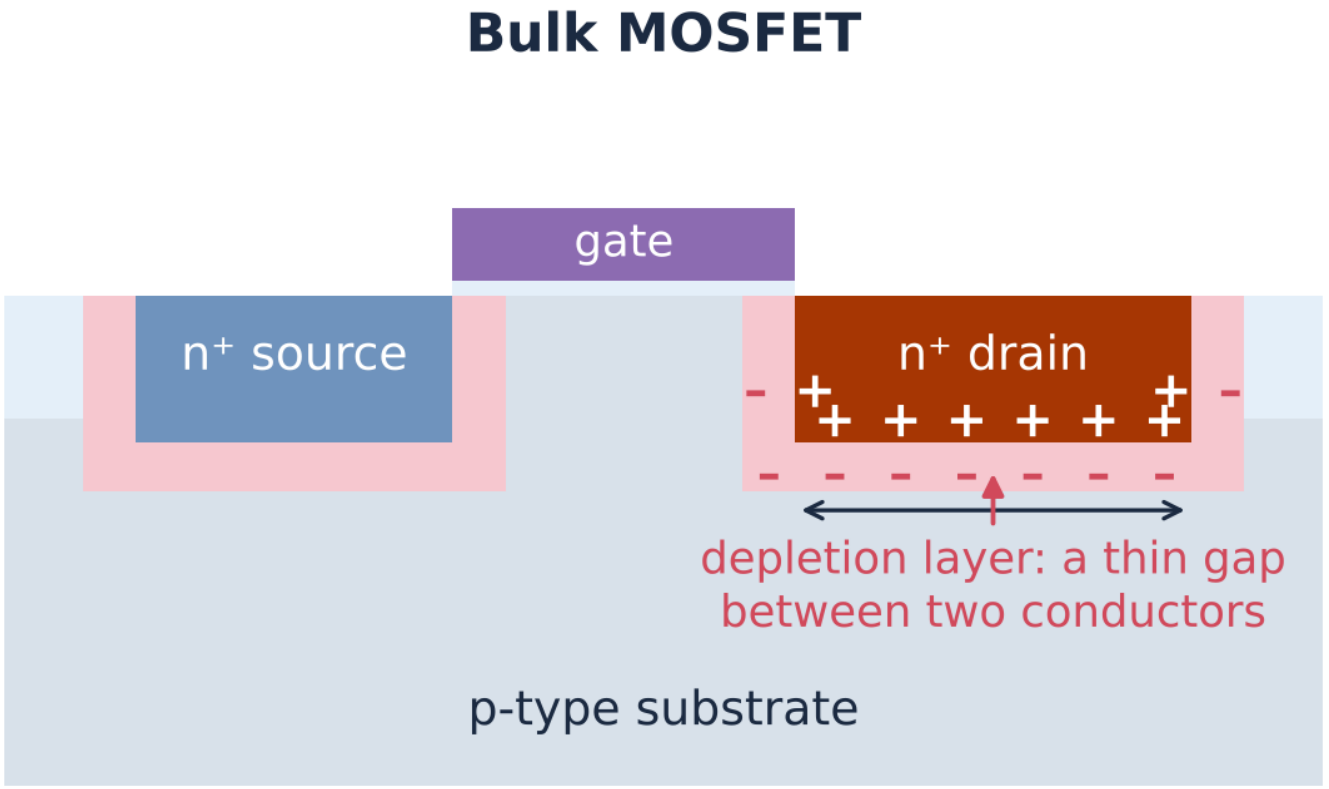


Figure made for this course.

SOI: put the drain on oxide

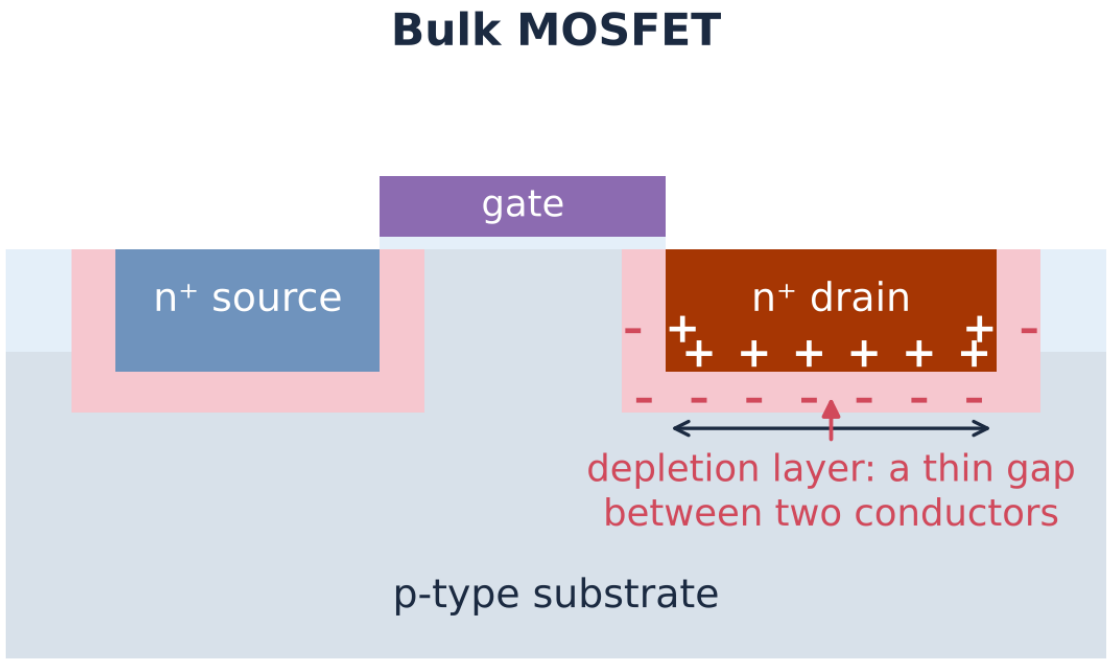


Figure made for this course.

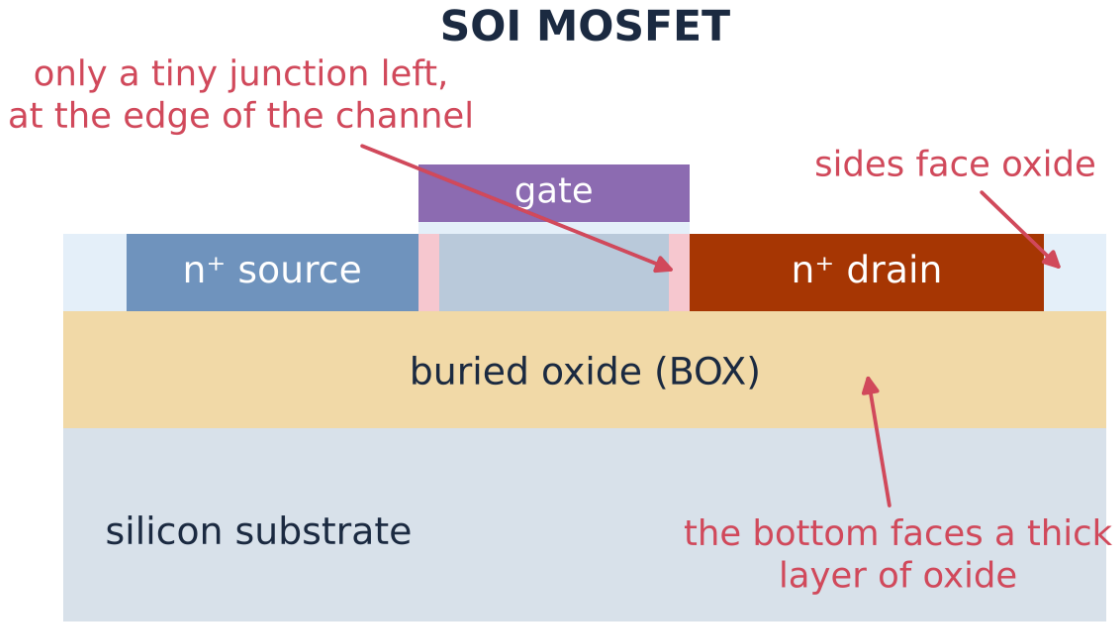
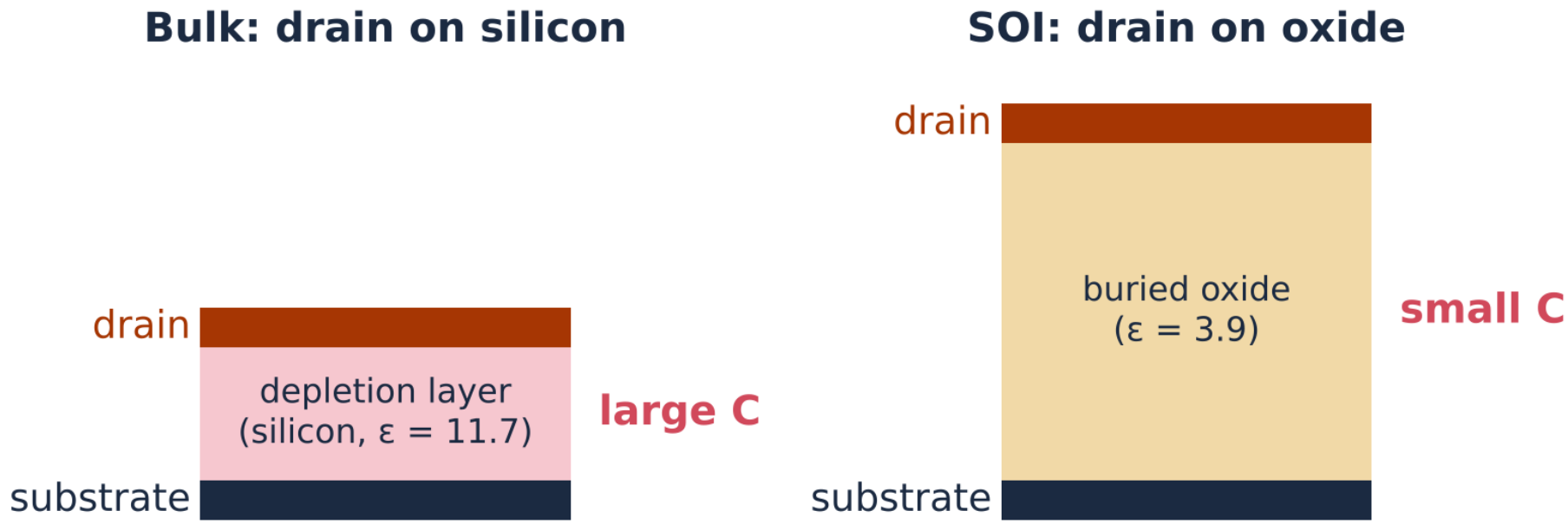


Figure made for this course.

In SOI the drain reaches down to the BOX. Its **bottom faces thick oxide**, its **sides face oxide**, and only a tiny junction is left at the edge of the channel.

Why the capacitance falls



C = ε A / d : a lower ε and a wider gap d both make C smaller (and the sides of the drain disappear too).

Figure made for this course.

A capacitor is **C = εA/d**. Under the drain, SOI swaps tens of nm of depleted silicon for **145 nm of oxide**, with **three times lower ε**; the side walls drop out too.

What a smaller capacitance buys

13.5 → 4.5 fF

the same 60 µm² drain at 2.5 V: bulk against 145 nm BOX (old notes' SPICE values)

1.2 → 0.7 V

supply needed for the same 40 GHz ring oscillator (2003 simulation)

266 → 66 µW

its power: four times less

Less capacitance to charge means the same speed at a **lower voltage**, and switching energy goes as **$C \cdot V^2$**. This is the main reason SOI was invented for fast, low-power chips.

A bonus: every transistor on its own island

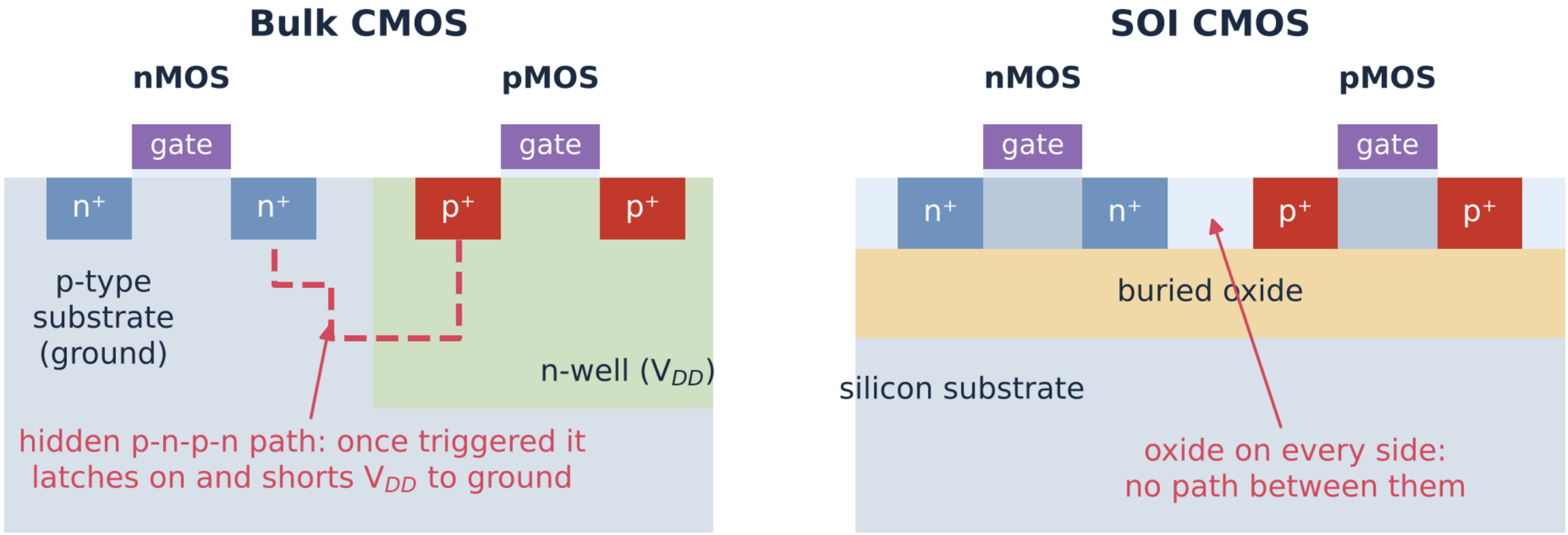


Figure made for this course.

Bulk CMOS hides a p-n-p-n path that can **latch up** and short the supply. In SOI, oxide surrounds each transistor: **no latch-up**, and much less charge collected from a cosmic ray.

3

Thick films and thin films

Does the depletion layer reach the oxide?

Partially or fully depleted?

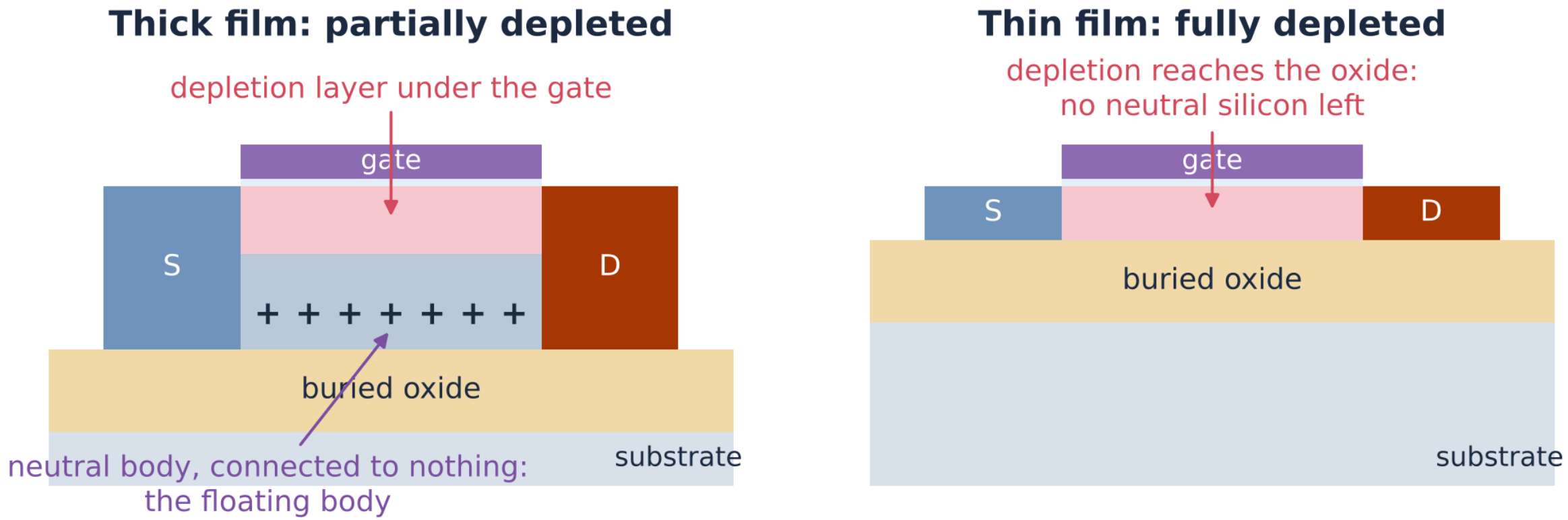
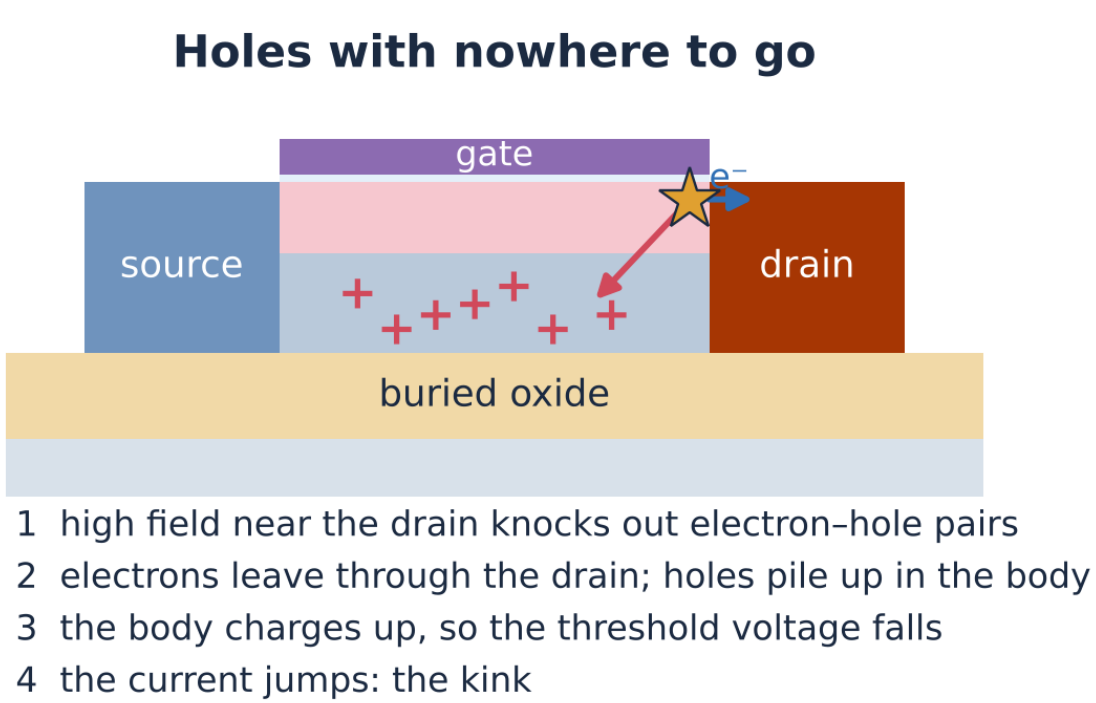


Figure made for this course.

In a **thick** film the gate depletes only the top; a neutral layer is left, connected to nothing. In a **thin** film the whole film is depleted.

Partially depleted: the floating body



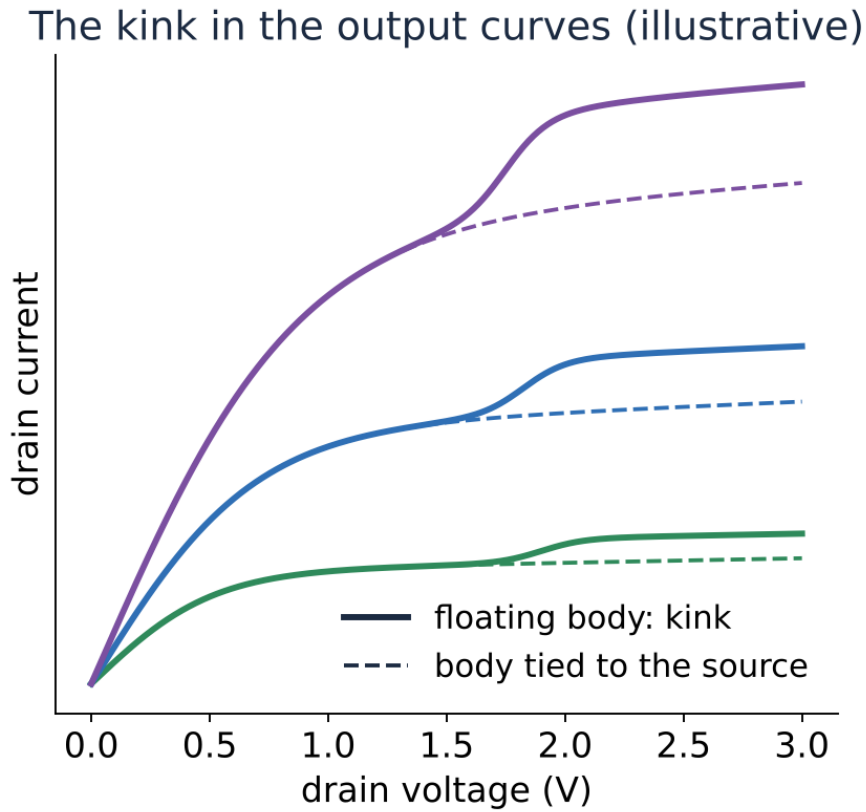
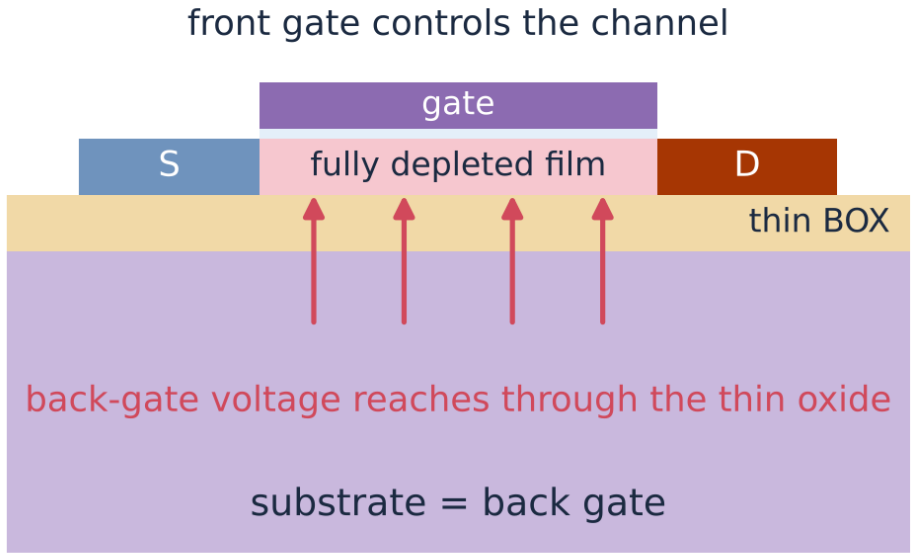


Figure made for this course.

Holes made near the drain have **nowhere to go**, so they charge the body and the current **jumps**: the kink. A **body tie** drains them, at a cost in area.

Fully depleted: a second gate for free



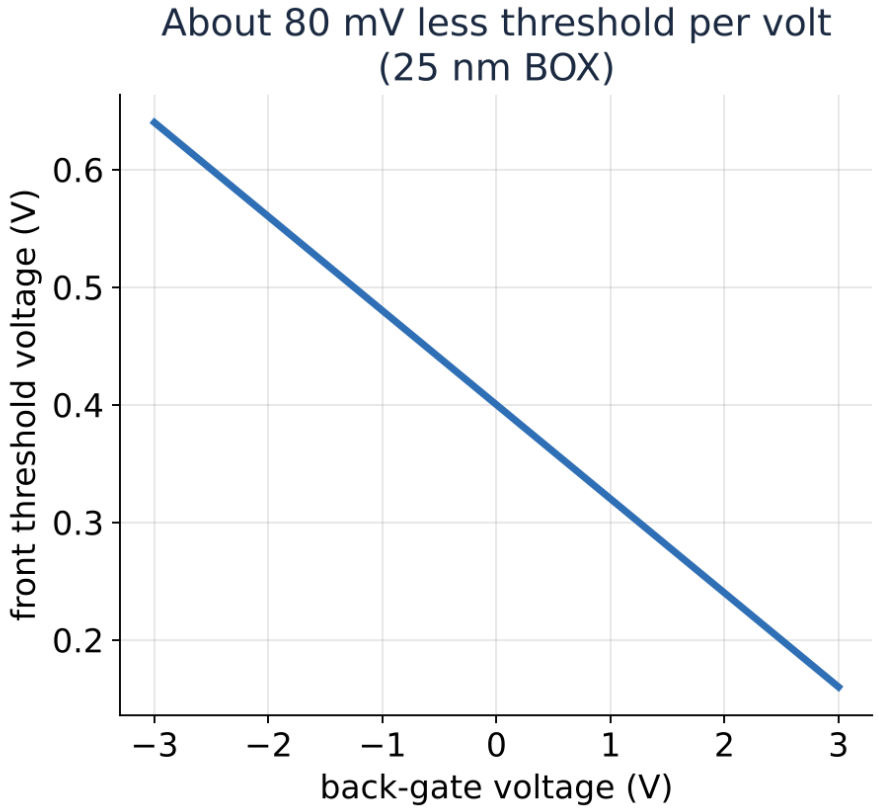


Figure made for this course; slope from the Lim–Fossum model.

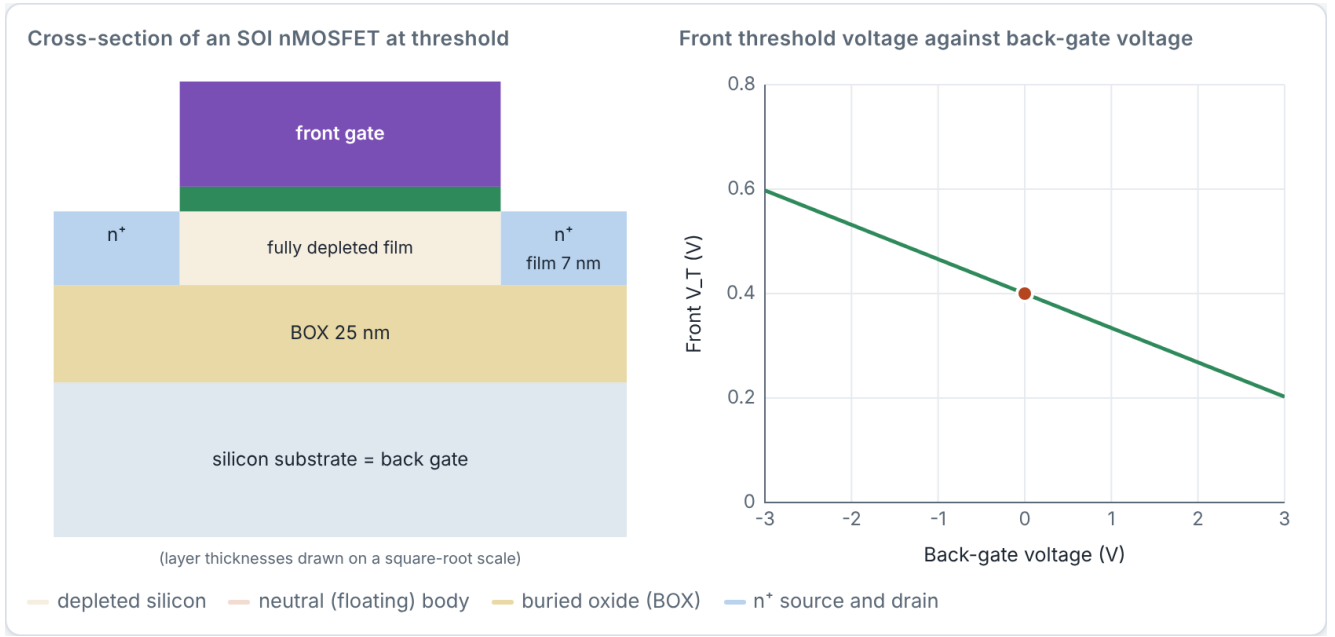
With no neutral layer, the substrate acts as a **back gate**. Its voltage tunes the threshold: **faster** when the chip is busy, **less leakage** when it idles.

Interactive animation 5

SOI: partially and fully depleted

faizverse.github.io/nanoelectronic-devices/L4/soi-pd-fd.html

- Thin the film from 150 nm until it becomes fully depleted
- 28 nm FD-SOI preset: sweep the back gate
- Make the BOX thick: what happens to the back gate's grip?
- More detail: find the kink, then add a body tie



Where SOI is used in 2026

FD-SOI

28, 22 and 18 nm: low-power chips, IoT, cars, radio.

RF-SOI

The antenna switches in almost every phone.

Photonics

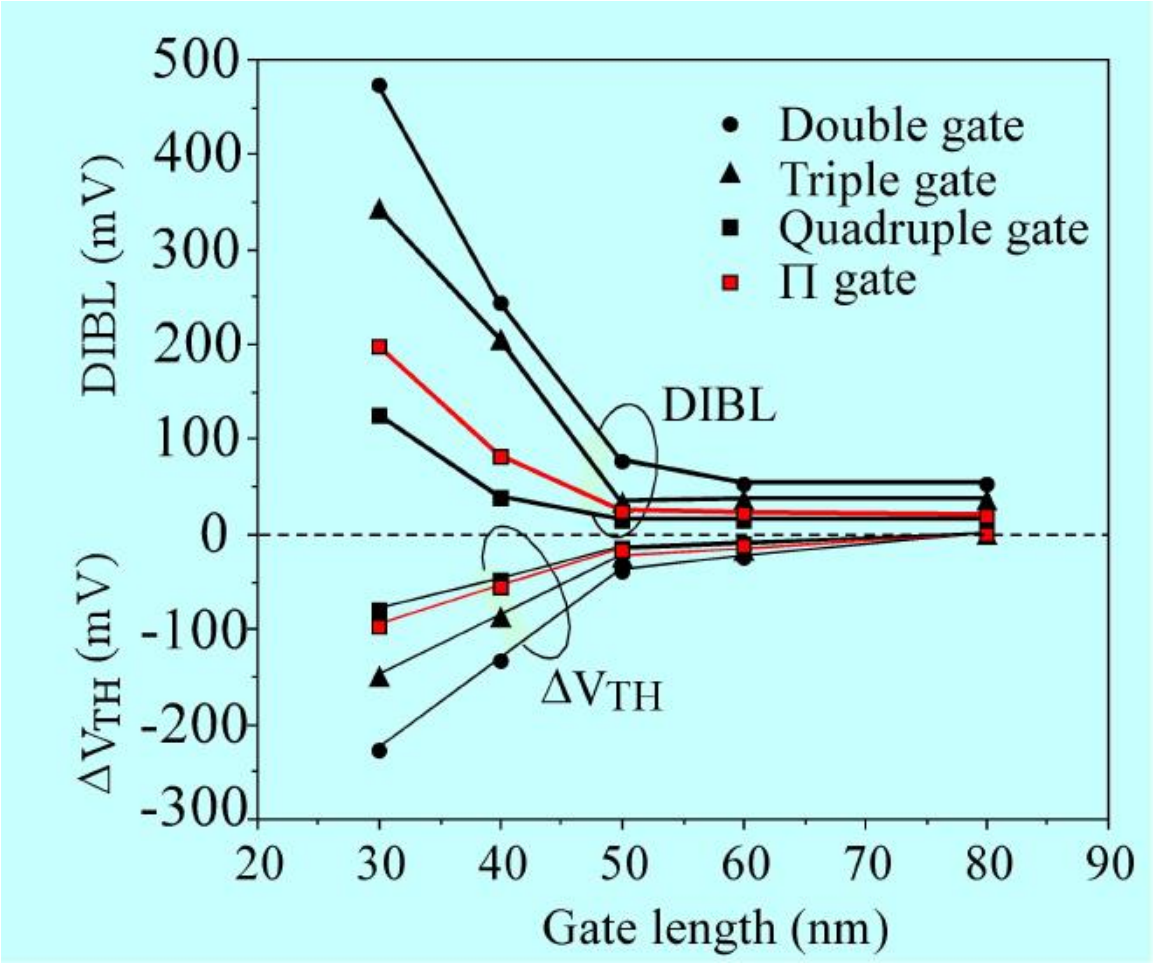
Light guided in silicon on the BOX: data-centre optical links.

Nanosheets

The logic end point: thin, fully depleted sheets with gate on all sides.

One idea, already in 2001

- Colinge compared SOI devices with **two, three and four** gates
- More gate around the channel: **less drain leakage** at every length
- SOI, the FinFET and the nanosheet are **one idea**: thin the body, surround it with gate



J.-P. Colinge, NSREC 2001 short course (old notes).

Part 2 in six points

1 The problem

In a short channel the drain reaches under the gate.

2 Two cures

Thin the body, or surround it with gate.

3 Fins and sheets

Taller fins and stacked sheets: more current on the same area.

4 The price

Oxide all round traps heat; resistance and stray capacitance grow.

5 SOI

The drain sits on oxide: about a third of the capacitance, so lower voltage and power.

6 PD and FD

Thick films float and kink; thin films get a back gate.

Credits

Figures

Drain-reach maps (computed), 3D device drawings, junction-capacitance, latch-up, PD/FD, floating-body, back-gate and heat pictures, fin-width chart and wafer steps made for KKKC6353 (scripts in figures-source). Reproduced from the old notes: DELTA (Hisamoto et al., 1990), the Berkeley FinFET (Yu, Hu et al., 2002) and spacer lithography (Choi et al., 2002), via Q. Zhang's EE 666 slides; multi-gate comparison from J.-P. Colinge's NSREC 2001 course; nanosheet process from IEEE Spectrum (2019) via the 2026 course slides.

Sources

Lim and Fossum, IEEE TED 30, 1244 (1983); Bruel (1995); Hisamoto et al., IEEE TED 47, 2320 (2000); Colinge, Solid-State Electronics 48, 897 (2004); Sicard and Delmas-Bendhia, Microwind ch. 15 (2003); K. N. Bhat, IISc lectures (2007); STMicroelectronics (2024). Full list in L4 Document 3.